

Diffusional coupling between microfractures and pore structure and its impact on nuclear magnetic resonance measurements in multiple-porosity systems

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ABSTRACT

Nuclear magnetic resonance (NMR) relaxation time measurements, although among the most accurate methods to estimate formation porosity, have been considered conventionally as insensitive to the presence of microfractures. Hence, the NMR responses in multiple-porosity systems, which may contain intergranular pores, microfractures, or channel-like inclusions, have not yet been thoroughly investigated. NMR pore-scale simulations using a random-walk algorithm enabled us to quantify the impact of microfractures/channels on NMR measurements and to propose a new concept of fracture-pore diffusional coupling in such heterogeneous systems. **We randomly distributed and oriented microfractures (or channels) in 3D pore-scale images of different rock matrices.** We then quantified the sensitivity of NMR T_2 (spin-spin relaxation time) distribution to the presence of microfractures (or channels) and compared the pore-scale simulation results against a previously published experimental study. The pore-scale simulation

results from synthetic rock samples revealed that NMR T_2 distribution can be influenced not only by the pore-size distribution but also significantly by fracture-pore diffusional coupling. The intergranular pore size can be underestimated by up to 29%, and the volume fraction of intergranular pores can be underestimated by more than 10%, if the impact of diffusional coupling was not taken into account in interpretation of NMR measurements. Furthermore, we developed a simplified 1D analytical model for fracture-pore diffusional coupling. The analytical solutions of the 1D model were in agreement with the simulation results in the synthetic rock samples, which further demonstrated the existence of fracture-pore coupling in multiple-porosity systems. The developed 1D model enabled real-time evaluation of diffusional coupling effect in the presence of microfractures and complex pore-size distribution. The results were promising for future applications of NMR relaxometry for the assessment of microfracture content, when combined with other conventional well logs.

INTRODUCTION

Assessment of pore structure within double-porosity carbonate formations has been possible by joint inversion of conventional well logs (Kazatchenko et al., 2007), but it remains challenging to evaluate the hierarchical structure of multiple-porosity systems. Borehole nuclear magnetic resonance (NMR) measurements have been among the most reliable techniques to evaluate porosity and pore-size distribution in various formations. There have been many petrophysical studies on NMR applications in porous media because the mechanism of proton relaxation in restricted space was

explained decades ago (Brownstein and Tarr, 1979; Kleinberg et al., 1994). However, there have been very limited studies on quantifying the effect of fractures in porous media on NMR measurements. The few published NMR studies on fractures only consider wide planar fractures, in which the saturating fluids behave like bulk fluids without spatial restrictions (Nakashima and Kikuchi, 2007; Xiao and Li, 2011). In other words, the fractures under study are considered as isolated from the porous structures of rock. In the first study, Nakashima and Kikuchi (2007) evaluate the NMR response to wide planar fractures and propose the use of borehole NMR measurements for assessment of fracture aperture. The following study by

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Xiao and Li (2011) systematically investigated the impact of fracture characteristics and tool parameters on NMR logs by theoretical analysis and numerical simulations. These studies assumed that free fluids exclusively occupy the fracture space and the fractures are parallel to each other and dominantly perpendicular to the wellbore. The findings from the aforementioned publications can only be applied to wide planar fractures with apertures much greater than 0.2 mm because a simple derivation shows that when the aperture is wider than 0.2 mm, the relaxation time associated to fractures approaches the bulk water relaxation time, i.e., 2 s, and becomes independent of aperture size (Nakashima and Kikuchi, 2007; Xiao and Li, 2011).

Furthermore, other publications studied the diffusional pore-to-pore exchange in nonfractured rock samples and their effect on NMR response. It has been demonstrated that intra- and intergranular pores in heterogeneous porous structures (e.g., carbonate rocks) can significantly influence the NMR T_2 distribution from each other. Ramakrishnan et al. (1999) propose an analytical model of diffusional pore coupling to describe the diffusion of water molecules between micro (intragranular) and macro (intergranular) pores in carbonate rocks. They observed that diffusional pore coupling can significantly decrease the longer relaxation time of macropores and suppress the peak amplitude of micropores that have a shorter relaxation time. Toumelin et al. (2002) further investigate the coupling between micro- and macroporosities in carbonate rocks and reported the same decrease by Monte Carlo simulations and experimental measurements. Anand and Hirasaki (2005) study the diffusional pore coupling in sandstones and grainstones and verify the decrease in macropore relaxation times by mathematical modeling and experimental results. A recent laboratory study (Grunewald and Knight, 2009) also confirmed the decrease in macropore relaxation times due to pore coupling by changing the surface geochemistry of porous materials.

None of these previous studies, however, take into account the microfractures in the rock. Thus, the diffusional coupling effect between microfractures and original rock pores has never been investigated. We hereby incorporate microfractures in porous media to study the diffusional coupling between microfractures and intergranular pores and show NMR sensitivity to the randomly distributed microfractures as well as the intra-/intergranular pores. Based on the results, we propose a new concept of fracture-pore diffusional coupling.

To substantiate our hypothesis that microfractures can influence the rock NMR signals, we start with a simple theoretical derivation. It is well known that, in the fast diffusion limit, the NMR spin-spin relaxation time T_2 of hydrogen nuclei in the fluids in porous media can be expressed as (Coates et al., 1999)

$$\begin{aligned} \frac{1}{T_2} &= \frac{1}{T_{2B}} + \frac{1}{T_{2S}} + \frac{1}{T_{2D}} \\ &= \frac{1}{T_{2B}} + \rho_2 \left(\frac{S}{V} \right)_{\text{pore}} + \frac{D(\gamma \cdot G \cdot TE)^2}{12}, \end{aligned} \quad (1)$$

where T_{2B} is the bulk relaxation time, T_{2S} is the surface relaxation time, T_{2D} is the diffusion-induced relaxation time, ρ_2 is the surface relaxivity, S/V is the surface-to-volume ratio of the pore space, D is the diffusion coefficient of pore fluids, γ is the gyromagnetic ratio of a proton, G is the internal magnetic field gradient, and TE is the interecho spacing time.

It has been well accepted that the NMR T_2 distribution represents the pore size distribution of the rock sample when (1) the pore space is fully saturated with fluids of high hydrogen index such as water or liquid hydrocarbon, (2) short TE in the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence is used, and (3) all the pore surface has a uniform surface relaxivity ρ_2 . Under these conditions the relaxation process is dominated by the surface relaxation mechanism, and the surface relaxation time T_{2S} can be considered as directly proportional to the pore size, given by

$$\frac{1}{T_{2S}} \propto \left(\frac{S}{V} \right)_{\text{pore}} \propto \left(\frac{1}{d} \right)_{\text{pore}}, \quad (2)$$

where d is the diameter of the pores. However, to validate the second part of equation 2, another assumption must be made; all the pores are similar in geometry, thus they possess similar surface-to-volume ratios.

For instance, let us consider the case when thin planar fractures exist in rock samples together with the spherical intergranular pores. The surface-to-volume ratios of spherical pores and planar fractures are given by

$$\left(\frac{S}{V} \right)_{\text{sphere}} = \left(\frac{6}{d} \right)_{\text{sphere}} \quad (3)$$

and

$$\left(\frac{S}{V} \right)_{\text{plane}} = \left(\frac{2}{w} \right)_{\text{plane}}, \quad (4)$$

where w is the thickness of planar fractures. We can expect that if the diameter of spherical pores is three times the thickness of planar fractures, they will overlap on T_2 distribution. In this case, if we still assume that the rock contains merely spherical pores, we may find mismatch between NMR T_2 distribution and pore size distribution of the rock.

The inherent assumption of similar pore geometry in equation 2 makes the impact of microfractures on NMR T_2 distribution not discernible. The purpose of this paper is therefore to evaluate (1) NMR sensitivity to the randomly distributed microfractures inside rocks and (2) the diffusional coupling effect between microfractures and intergranular pores in various formations including sandstones and carbonates.

The following sections describe the methods, the results of numerical simulations in sandstone and carbonate core samples as well as the theoretical analysis to confirm the simulation results.

METHODS

The numerical method starts with generation of digital pore-scale images of different rock samples, followed by simulations of NMR response in these 3D pore-scale images. Finally, we describe a 1D analytical model for NMR diffusional coupling.

Pore-scale images of multiple-porosity structures

The pore-scale images, as the input to the NMR simulator, can be obtained by micro-computed tomography (CT) or focused ion beam scanning electron microscope (FIB-SEM) depending on the rock

type and pore size. We use Fiji, an open-source image processing software (Schindelin et al., 2012), to convert the original grayscale micro-CT images to binary (black-and-white) images by trainable segmentation. The main idea of trainable segmentation is to train the software to differentiate pores and grains by manually drawing lines on one of the micro-CT images, then the software apply this segmentation algorithm to all the other images. By this means, the noise pixels on images are eliminated. Next, we use ImageJ, a public-domain, Java-based image-processing program developed at the National Institute of Health to reconstruct a 3D rock matrix from the binary images. We then synthetically create randomly distributed and oriented microfractures with different size, concentration, and shape (e.g., planar fractures or channel-like inclusions). Finally, we incorporate the synthetic microfractures into the 3D matrices of carbonate/sandstone rock samples. All the synthetic cases are matrices of 0 s and 1 s (0 represents pore pixel, and 1 represents grain pixel), with $300 \times 300 \times 300$ pixels in volume.

Nuclear magnetic resonance pore-scale simulations and T_2 inversion

The NMR magnetization decay is simulated using a random-walk algorithm (Ramakrishnan et al., 1998; Øren and Bakke, 2002) via a C++ program adapted from the work of Talabi et al. (2009). In all the simulations, we assume that the pore space is fully saturated with brine (i.e., T_{2B} of the saturating fluid is assumed to be 3.1 s), and there is no diffusion-induced relaxation (i.e., G is zero in equation 1). For the sandstone and carbonate examples, the image resolution is assumed to be $1 \mu\text{m}/\text{pixel}$. The number of walkers assigned in each pore voxel is either 1 or 4, depending on the pore size. The surface relaxivity of carbonate rocks typically ranges from 1 to $10 \mu\text{m}/\text{s}$ (Talabi, 2008), and for sandstone rocks it varies from 9 (Liaw et al., 1996) to $46 \mu\text{m}/\text{s}$ (Roberts et al., 1995). Thus, we assume a surface relaxivity of $10 \mu\text{m}/\text{s}$ for sandstone and carbonate rocks for convenience of comparison. In this paper, we do not investigate the influence of surface relaxivity change on the NMR response. We expect that with higher surface relaxivities, the impact of diffusional coupling on NMR response decreases (Ramakrishnan et al., 1999).

The NMR T_2 distribution is inverted from the magnetization decay by using a curvature smoothing method (Chen et al., 1999; Toumelin et al., 2003; Talabi et al., 2009). The area under each T_2 distribution curve represents the total porosity of the rock.

Analytical model for diffusional coupling

The 1D analytical model for the diffusional coupling effect is adapted from a previous model introduced by Ramakrishnan et al. (1999). The original 1D model described the pore coupling between macropores (intergranular) and micropores (intragranular pores) in carbonate rocks. We adapted this model to study the diffusional coupling between microfractures and intergranular pores in various rock samples. Figure 1 shows the assumed 1D simplified model for the fractured rock sample. The magnetization in the

fracture domain M_F is governed by the usual NMR diffusion equation:

$$D \frac{\partial^2 M_F}{\partial x^2} - \frac{M_F}{T_{2B}} = \frac{\partial M_F}{\partial t}, \quad (5)$$

where D is the diffusion coefficient of fluids within the fractures.

Within the porous grains, the magnetization M_G is governed by the modified NMR diffusion equation:

$$\frac{D}{F} \frac{\partial^2 M_G}{\partial x^2} - \phi_G M_G \left(\frac{1}{T_{2B}} + \frac{1}{T_{2,G}} \right) = \phi_G \frac{\partial M_G}{\partial t}, \quad (6)$$

where $T_{2,G}$ is the relaxation time of fluids within the intergranular pores, and F and ϕ_G are the formation factor and the intergranular porosity of the porous grain domain, respectively.

The boundary conditions between the fracture domain and the porous grain domain can be expressed as

$$M_F|_{x=H} = M_G|_{x=H} \quad (7)$$

and

$$D \frac{dM_F}{dx} \Big|_{x=H} = \frac{D}{F} \frac{dM_G}{dx} \Big|_{x=H} - \rho(1 - \phi_G) M_F|_{x=H}, \quad (8)$$

where ρ is the surface relaxivity of the grains, and H is half the thickness of fractures given by $H = w/2$.

According to Ramakrishnan et al. (1999) the approximate analytical solutions to equations 5 and 6 can be obtained by assuming two principal exponential-decay modes. First, we define *coupled fractures* as the microfractures that are diffusionally coupled (i.e., connected) with intergranular pores. Next, we adopt Ramakrishnan et al.'s (1999) approximate analytical solution to obtain the relaxation time for the coupled fractures $T_{2F,\text{coupled}}$ via

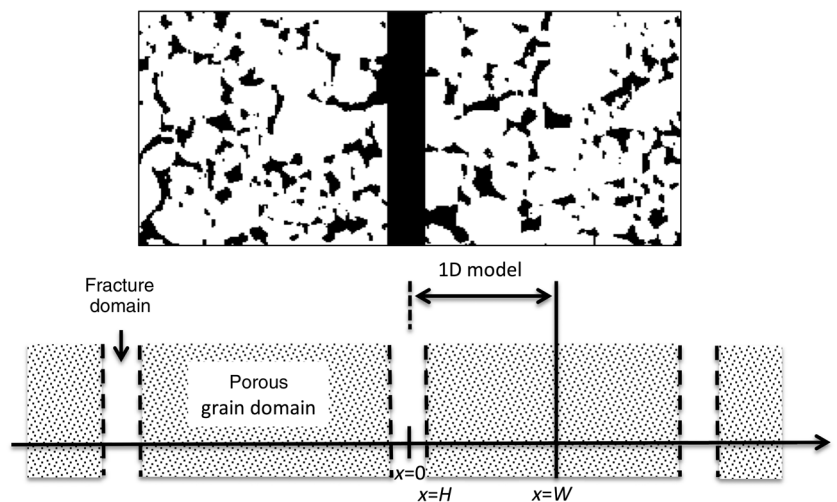


Figure 1. Simplified 1D model for a fractured rock matrix. The values on the x -axis are used to denote the relative dimensions of fractures and grains. H is half the thickness of fractures, and W is the distance from fracture center to grain center.

$$\frac{1}{T_{2F,coupled}} \cong \frac{1}{H} \left[\sqrt{\frac{\phi_G D}{T_{2,G} F}} + \rho(1 - \phi_G) \right]. \quad (9)$$

where

$$X \cong \frac{T_{2,G}/T_{2,F,coupled}}{\sqrt{1 - T_{2,G}/T_{2,F,coupled}}}. \quad (12)$$

Equations 5–9 are adapted from Ramakrishnan et al. (1999) by replacing their terms for micro- and macropores with our definitions of intergranular pores and microfractures, respectively.

In comparison, if the microfractures are not connected to any intergranular pores, we define them as *isolated fractures*. The relaxation time of fluids within the isolated fractures $T_{2F,iso}$ can be theoretically estimated by

$$\frac{1}{T_{2F,iso}} = \rho \cdot \frac{2}{w} = \frac{\rho}{H}. \quad (10)$$

It can be easily shown that when the grain porosity ϕ_G approaches zero, equation 9 naturally becomes equation 10, which describes isolated microfractures.

The amplitude of T_2 mode in the isolated microfractures I_{iso} can be determined from the input geometry. The amplitude of T_2 mode in the coupled microfractures $I_{coupled}$ is then derived using the analytical model, given by (Ramakrishnan et al., 1999)

$$I_{coupled} \cong I_{iso} \frac{(1 + X)^2}{1 + X/2}, \quad (11)$$

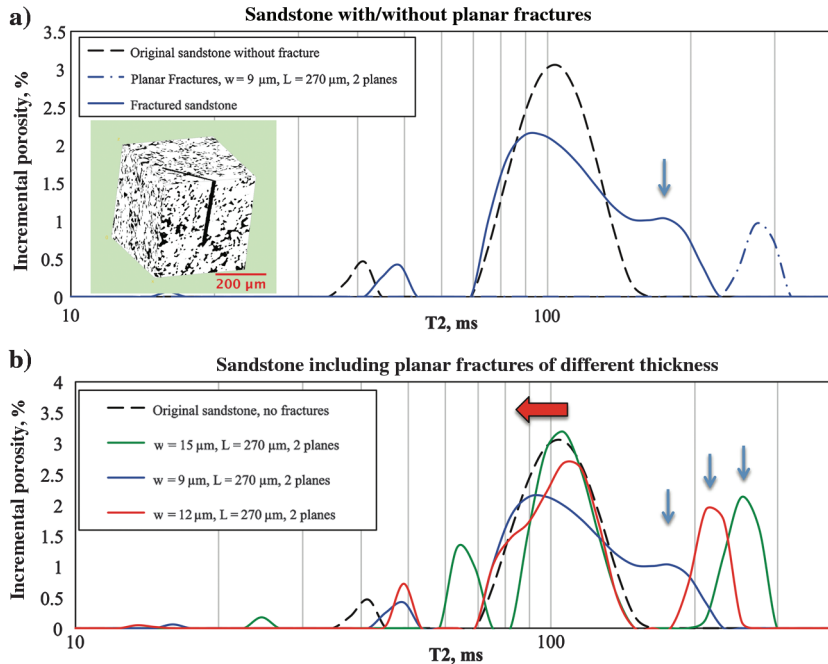


Figure 2. Sandstone rock sample: Simulated NMR T_2 distribution (a) in the rock with two planar fractures (solid blue line) with width (w) of 9 μm and length (L) of 270 μm and without planar fractures (black dashed line). Blue dot-dashed line shows the simulated NMR T_2 distribution in the presence of only planar fractures. The blue arrows denote the T_2 peak corresponding to the planar fractures, which are diffusionally coupled with the pore system. The 3D pore-scale image on the left-hand side shows the sandstone example with two 9- μm planar fractures and (b) in the rock including two planar fractures with width (w) of 9, 12, or 15 μm and length (L) of 270 μm . Three blue arrows denote the T_2 peaks for the planar fractures that are diffusionally coupled with the pore system. The area under each T_2 distribution curve represents the total porosity of the rock.

SYNTHETIC CASE NO. 1: SANDSTONE ROCK SAMPLE

Synthetic case no. 1 is designed to illustrate the impact of microfractures in a porous media with highly connected intergranular pore structure such as conventional sandstone reservoirs. We synthetically generate pore-scale images of fractured sandstone rocks, by inserting synthetic planar fractures into the sandstone matrix. The base micro-CT image of the sandstone rock sample is provided by the Pore-scale Modeling research group at Imperial College London (Dong, 2007). We assign four walkers in each pore voxel for NMR random-walk simulations.

Figure 2a shows the NMR T_2 distribution for the synthetic sandstone rock sample including and in the absence of planar fractures. The same figure also shows NMR T_2 distribution for the fractures isolated from the sandstone rock matrix. The two square planar fractures in the sandstone rock sample are identical in shape and size, with thickness of 9 μm and length of 270 μm . The results show that the relaxation times associated with the isolated planar fractures (blue dashed line) are longer than those of the original sandstone matrix (black dashed line). This is expected because the thickness of planar fractures, 9 μm , is larger than one-third of the dominant pore size in the sandstone rock sample, which is estimated as 14.9 μm . We also calculated the T_2 value of the isolated fractures analytically (Appendix A). The results from analytical calculations confirm the simulation results for isolated fractures.

After incorporating the fractures into the sandstone matrix, the resulting T_2 distribution (blue solid line) is not equivalent to the superposition of the previously presented T_2 distributions (blue and black dash lines). Instead, the relaxation time corresponding to the planar fractures has been significantly shortened from 274 (isolated fractures) to 178 ms (coupled fractures). Meanwhile, the pore relaxation time has been slightly decreased. The T_2 values for intergranular pores, coupled fractures, and isolated fractures are listed in Table 1.

This significant decrease in the fracture T_2 relaxation time and the slight decrease in the pore relaxation time can be explained by the concept of diffusional coupling proposed by Ramakrishnan et al. (1999). The physical origin of the decrease in longer T_2 is that, when the water molecules originally in large pores diffuse into small pores in which they are exposed to more surfaces, their proton spins relax faster. This effect is obvious when the fluid diffusion rate is faster than the characteristic relaxation rate of pores (Ramakrishnan et al., 1999; Anand and

Hirasaki, 2005; Grunewald and Knight, 2009). We need to point out that the slight decrease in shorter T_2 times observed in our simulation results (Figure 2a) was also predicted by Ramakrishnan et al.'s (1999) theoretical analysis, but it was rarely mentioned or explained in literature. In conclusion, we demonstrate that the fracture-pore diffusional coupling exist in this sandstone example. Therefore, the diffusional pore-coupling concept, although proposed for bimodal pores in carbonate rocks, can be extended to explain NMR responses in fractured rocks.

Figure 2b shows the impact of fracture thickness on NMR T_2 distribution. The results show that shrinking fracture thickness leads to a more obvious decrease in sandstone pore relaxation times. When the fractures narrow down from 15, 12, to 9 μm , the pore T_2 slightly shifts downward (i.e., from 106.3 to 97.5 ms, denoted by a red horizontal arrow). Specifically, in the case of 12- μm fractures (red line), the pore T_2 starts shifting downward as shown by the shoulder peak on the left side. The reason is that the fracture thickness gets closer to one-third pore size, the diffusional coupling among them gets stronger, thus showing a more obvious pore T_2 decrease. Indeed, the amount of exchanged fluid between fractures and pores becomes nonnegligible when their sizes are similar, which promotes the fracture-pore coupling. Furthermore, Chang et al. (1997) publish experimental results of real limestone samples including artificial fractures and show the same decrease in relaxation times of limestone pores. Chang et al. (1997) provide T_2 distribution results for four Leuters limestone samples, three out of which were made with artificial fractures with controlled widths of 0.75, 0.35, and 0.25 mm, respectively. Their results show a visible decrease in the limestone relaxation times due to the existence of artificial fractures, in water- and oil-saturated cases (i.e., from ~ 3.1 to ~ 2.5 ms in water-saturated case). Although the authors did not mention or explain this phenomenon, we find their experimental results in agreement with our simulation results and Ramakrishnan et al.'s (1999) analytical solutions. Again, these evidences demonstrate the existence of fracture-pore coupling effect in the fractured rocks.

Furthermore, we estimate the volume occupied by the intergranular pores and the fractures, respectively, from NMR T_2 distribution of our sandstone example. We calculate the volumes by integrating the area below the T_2 curve that represents pores or fractures. In the 9- μm fracture case in which the pore and fracture peaks

are merged together, we select the saddle point (163 ms) as the T_2 cutoff value between the fractures and pores, then we integrate the areas above and below the T_2 cutoff value to calculate volumes of fractures and pores, respectively. The results in Table 1 show that the volume ratio of pores versus fractures (i.e., pore/fracture volume ratio) is underestimated by NMR T_2 distribution, compared with the value calculated from the input images, for all three sandstone cases. This result is further substantiated by Ramakrishnan et al.'s (1999) observation that the amplitude of short T_2 peak gradually decreases when coupled with the long T_2 peak. This observation indicates that the volume fraction of the short T_2 component (i.e., intergranular pores in this case) might be underestimated by more than 10% in NMR T_2 distribution when diffusional coupling exists.

SYNTHETIC CASE NO. 2: CARBONATE ROCK SAMPLE

This set of synthetic carbonate examples is designed to compare the impact of microfractures in carbonate rock samples with distinct pore-size distributions. We synthetically generate two pore-scale images of fractured carbonate rocks: one with small pores and the other one with large pores. The dominant intergranular pore sizes are estimated as 10 and 57 μm , for the small-pore and large-pore carbonate rock samples, respectively. The base micro-CT images of the carbonate rock examples are provided by a project funded by Qatar National Research Fund (Gupta et al., 2010). The synthetic microfractures are either planar fractures or channel-like inclusions.

Small-pore carbonate example including planar fractures

This carbonate example illustrates the impact of microfractures in a porous media with poorly connected intergranular pore structure. The volumetric Euler-Poincaré characteristic (EPC) method is used to quantify the pore connectivity in porous media (Vogel, 1997; Chi and Heidari, 2014), with lower value of EPC representing better pore connectivity. We calculate the volumetric EPC for the carbonate and the sandstone samples, which are -1.38×10^{-4} and $4.37 \times 10^{-4} \mu\text{m}^{-3}$, respectively. It shows that the carbonate sample has poorly connected pores compared with the well-connected sandstone sample. The same sets of planar fractures, as used in

Table 1. Sandstone rock sample: Calculated total porosity, pore T_2 value, coupled and isolated fracture T_2 values, fracture T_2 relative decrease compared with isolated fractures, and pore/fracture volume ratios calculated from images and estimated from NMR T_2 distributions, for the sandstone rock shown in Figure 2.

Matrix	Sandstone, no fractures	Presence of 9- μm fractures	Presence of 12- μm fractures	Presence of 15- μm fractures
Porosity, %	16.86	18.57	19.15	19.73
Pore T_2 , ms	106.3	97.5	106.3	106.3
Coupled fracture T_2 , ms	—	178	212	251
Isolated fracture T_2 , ms	—	274	354	421
Fracture T_2 decrease, %	—	35	40	40
Pore/fracture volume ratio (images)	—	8.01	5.97	4.75
Pore/fracture volume ratio (NMR)	—	5.63	3.25	2.88

the sandstone example (i.e., two identical planes with width of 9, 12, or 15 μm and length of 270 μm), are inserted into the carbonate matrix. We assign four walkers in each pore voxel for NMR random-walk simulations.

Figure 3 shows the impact of fracture thickness on NMR T_2 distribution in the carbonate case. Table 2 lists the T_2 values estimated for intergranular pores, coupled fractures, and isolated fractures. We observe that, unlike the sandstone case, shrinking the fracture thickness has no impact on the carbonate pore relaxation times; i.e., the pore T_2 decrease is indiscernible. Meanwhile, the fracture T_2 decrease is less than that in the sandstone sample. We observe that in the carbonate case the coupled fracture T_2 are decreased by 29% (Table 2) relative to the isolated fracture T_2 , which is smaller than the 35% ~ 40% decrease in the sandstone case (Table 1). Total porosity and intergranular pore sizes are similar for the carbonate and sandstone samples. However, the sandstone pore connectivity is much better than that in the carbonate sample. Consequently, we conclude that in the highly connected pore structure, the fracture T_2 is shortened more significantly than in the poorly connected pore structure. We attribute this observation to the stronger fracture-pore coupling in the better-connected pore structure. Indeed, in a highly connected pore space, the fluid molecules can migrate between fractures and pores more easily, which promotes the diffusional coupling between them.

Small-pore carbonate example including channel-like inclusions

This example illustrates the impact of microinclusions with apertures close to the size of intergranular pores, on NMR T_2 distribution. Channel-like inclusions, with radius of 3 μm and length of

150 μm , are inserted into the carbonate rock as a special form of microfractures in this example. It can be shown that channel-like inclusions and planar fractures have similar impact on NMR T_2 distribution by comparing equation A-1 with A-3 in the Appendix A. The relaxation times for both of them are solely affected by the size of their smallest dimension, which is the diameter of a cylindrical channel or the thickness of a planar fracture. Consequently, all the conclusions previously drawn for planar fractures can be applied to channel-like inclusions. We assign four walkers in each pore voxel for NMR random-walk simulations in this synthetic example.

Figure 4 shows the impact of channel-like inclusions on NMR T_2 distribution of the carbonate example. We observe an overlap of T_2 peaks for channels and intergranular pores (both located at 106.3 ms). Although the amplitude of T_2 peak increases with higher concentration of inserted channels, the channels and intergranular pores cannot be distinguished from the T_2 distribution in this case. Table 3 shows the calculated total porosity, pore T_2 value, and coupled channel T_2 values for the carbonate example.

The major difference between planar fractures and channel-like inclusions is that, given the same surface-to-volume ratio, a planar fracture occupies a larger volume than a cylindrical channel, so the NMR T_2 distribution is more sensitive to planar fractures. In other words, planar fractures can be considered as expanded cylindrical channels.

Large-pore carbonate example including planar fractures

In this example, we use a large-pore carbonate matrix to study the effect of thin microfractures on large pores. In this synthetic case, the total volume of microfractures are kept the same, whereas the

Figure 3. Small-pore carbonate rock sample: Simulated NMR T_2 distribution for the synthetic carbonate example with planar fractures with width (w) of 9, 12, or 15 μm and length (L) of 270 μm . Three blue arrows denote the T_2 peaks for the planar fractures, which are diffusionally coupled with the pore system. The 3D pore-scale image on the left-hand side shows the carbonate example with two 15- μm planar fractures. The area under each T_2 distribution curve represents the total porosity of the rock.

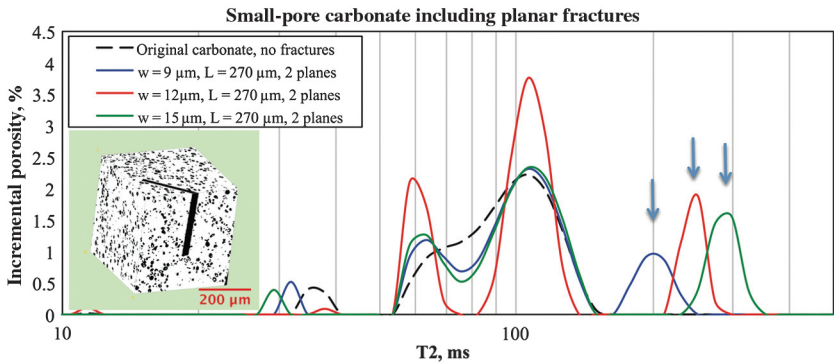


Table 2. Small-pore carbonate rock sample: Calculated total porosity, pore T_2 value, coupled and isolated fracture T_2 values, and fracture T_2 relative decrease compared with isolated fractures, for the sandstone rock shown in Figure 3.

Matrix	Small-pore carbonate, no fractures	Presence of 9- μm fractures	Presence of 12- μm fractures	Presence of 15- μm fractures
Porosity, %	15.70	17.44	18.02	18.60
Pore T_2 , ms	106.3	106.3	106.3	106.3
Coupled fracture T_2 , ms	—	194	251	298
Isolated fracture T_2 , ms	—	274	354	421
Fracture T_2 decrease, %	—	29	29	29

impact of fracture thickness on NMR T_2 distribution is evaluated. Square planar fractures with different thickness (i.e., 3, 6, or 9 μm) and the same length (150 μm) are inserted into the large-pore carbonate matrix. The aspect ratios (thickness-to-length ratio) of these fractures change from 0.02 to 0.04 to 0.06. The number of fractures is varied (i.e., 60, 30, or 20 fractures) to maintain the same total porosity. We assign one walker in each pore voxel for NMR random-walk simulations in this case.

Figure 5 shows the simulation results for different fracture sizes. Three blue arrows denote the T_2 peaks of the planar fractures. By theoretical calculation (see Appendix A) for the 3- μm fractures and separate numerical simulation on a cubic volume containing only

the 3- μm fractures, we determine that the 3- μm fractures are located at $T_2 = 107$ ms. Also 6- and 9- μm fractures are determined and labeled in the same way. The second blue T_2 peak ($w = 3$ μm case) in the middle is caused by original carbonate pores, which has the same origin as the small black peak. This middle peak disappears in the red (6 μm) and green (9 μm) cases because it merges with the fracture T_2 peaks.

We observe that in the presence of thin planar fractures, relaxation times of the large intergranular pores are more significantly decreased compared with the case of thick fractures. For example, the third blue T_2 peak in the $w = 3$ μm case represents the carbonate intergranular pores. When no fractures exist, this carbonate pore

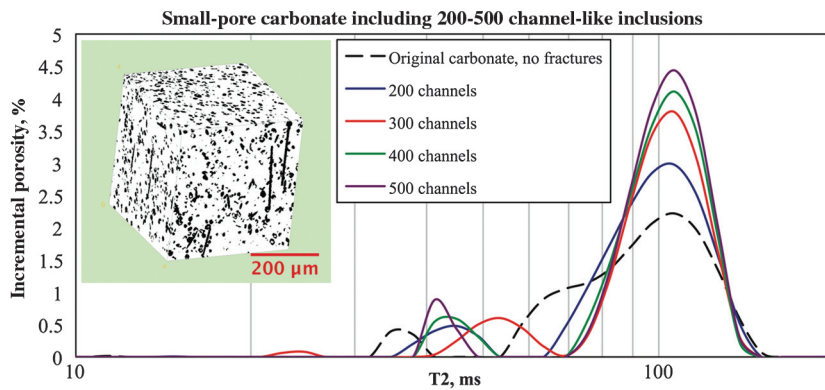


Figure 4. Small-pore carbonate rock sample: Simulated NMR T_2 distribution for the carbonate example including 200–500 channel-like inclusions. The 3D pore-scale image on the left-hand side shows the carbonate pore structure including 500 channels. The area under each T_2 distribution curve represents the total porosity of the rock.

Table 3. Small-pore carbonate rock sample: Calculated total porosity, pore T_2 value, and coupled channel T_2 values for the carbonate rock shown in Figure 4.

Matrix	Small-pore carbonate, no fractures	Presence of 200 channels	Presence of 300 channels	Presence of 400 channels	Presence of 500 channels
Porosity, %	15.70	17.47	18.35	19.25	20.02
Pore T_2 , ms	106.3	106.3	106.3	106.3	106.3
Coupled channel T_2 , ms	—	106.3	106.3	106.3	106.3

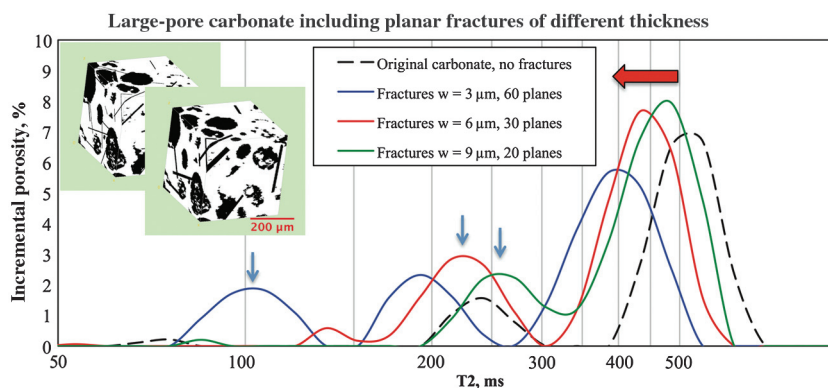


Figure 5. Large-pore carbonate rock sample: Simulated NMR T_2 distribution for the synthetic large-pore carbonate example, in presence of planar fractures of different widths. The planar fractures take width (w) of 3, 6, or 9 μm and keep the same length (L) of 150 μm . The number of planar fractures is 60, 30, or 20 for the three cases, respectively, to maintain a constant total porosity. Three blue arrows denote the T_2 peaks representing the planar fractures that are coupled with the pore system. The red arrow indicates that thin fractures can shorten the pore T_2 more than thick fractures. The 3D pore-scale images on the left-hand side show the carbonate pore structure including planar fractures with widths of 3 (left) and 9 μm (right image). The area under each T_2 distribution curve represents the total porosity of the rock.

peak is located where the largest black peak is; after fracturing, the peak is shifted downward by fracture-pore coupling. The pore T_2 peak is shifted downward by 11%, 21%, and 29% from its original position in the presence of 9-, 6-, and 3- μm planar fractures, respectively (Table 4). Again, this is explained by the fluid diffusion between fractures and pores — when the water molecules diffuse from large pores to thin fractures, they experience a larger surface area that makes the proton spins relax faster. The thinner fractures have larger surface-to-volume ratio, thus they relax the protons originally from large pores faster than thicker fractures. In other words, the relaxation time of the long-time component reflects an averaging of the sizes of heterogeneous porous structures when diffusional coupling exists.

Large-pore carbonate example including channel-like inclusions

In this synthetic case, the same sets of channel-like inclusions with radius of 3 μm and length of 150 μm , as used in the small-pore carbonate example, are inserted into the large-pore carbonate rock. We assign one walker in each pore voxel for NMR random-walk simulations.

Figure 6 shows the impact of volumetric concentration of micro-inclusions on NMR T_2 distribution in the large-pore carbonate sample. Table 5 lists the T_2 values estimated for intergranular pores. We observe that when thin channel-like fractures are inserted into the carbonate rock, the upper peak associated with the intergranular pores is shifted downward. Having more channels inserted into the carbonate rock, the upper peak shifts more significantly (i.e., denoted by the red arrow). Table 5 lists the estimated pore T_2 values and their corresponding decrease due to the increase in channels.

The results indicate that in the presence of higher volumetric concentration of microfractures, the fracture-pore coupling becomes stronger, causing the fluids in the large pores relax faster. In this case, we will significantly underestimate the pore size based on NMR T_2 distribution by up to 14%, if we ignore the microfractures (channel-like inclusions) and assume only spherical intergranular pores in the rock.

COMPARISON OF SIMULATION RESULTS AGAINST ANALYTICAL MODEL

We further compare our simulation results against the 1D analytical solution described in the method section. Table 6 lists the assumed parameters for the analytical model. We assumed a typical

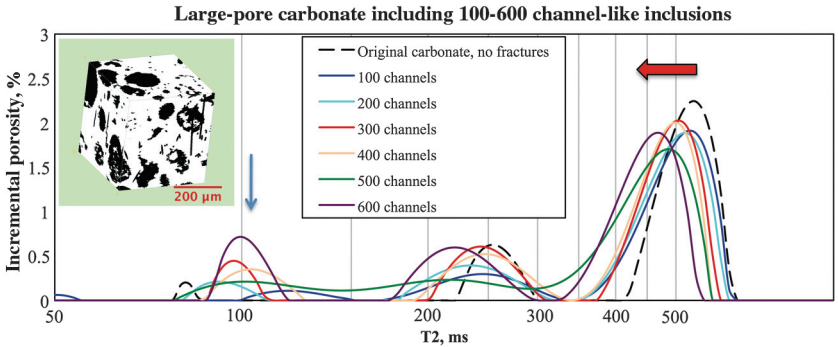
Table 5. Large-pore carbonate rock sample: Calculated total porosity and pore T_2 values for the large-pore carbonate example in Figure 6.

Matrix	Porosity, %	Pore T_2 , ms
Large-pore carbonate, no fractures	21.98	540.1
Presence of 100 channel-like fractures	22.84	523.7
Presence of 200 channel-like fractures	23.63	523.7
Presence of 300 channel-like fractures	24.47	507.8
Presence of 400 channel-like fractures	25.29	507.8
Presence of 500 channel-like fractures	25.99	492.4
Presence of 600 channel-like fractures	27.74	463.0

Table 4. Large-pore carbonate rock sample: Calculated total porosity, pore T_2 value, pore T_2 relative decrease compared with the original value, and coupled fracture T_2 values for the large-pore carbonate rock example shown in Figure 5.

Matrix	Large-pore carbonate, no fractures	Presence of 3- μm fractures (60 fractures)	Presence of 6- μm fractures (30 fractures)	Presence of 9- μm fractures (20 fractures)
Porosity, %	21.98	31.11	30.96	30.96
Pore T_2 , ms	546.1	385.3	432.8	486.1
Pore T_2 decrease, %	—	29	21	11
Coupled fracture T_2 , ms	—	107.2	215.4	271.8

Figure 6. Large-pore carbonate rock sample: Simulated NMR T_2 distribution for the synthetic large-pore carbonate example including 100–600 channel-like inclusions. The red arrow denotes that pore T_2 decreases as more fractures are inserted into the matrix, and the blue arrow indicates the fracture T_2 location at around 100 ms. The 3D pore-scale image on the left-hand side shows the large-pore carbonate sample including 500 channels. The area under each T_2 distribution curve represents the total porosity of the rock.



formation factor of 25 for the sandstone example (Ramakrishnan et al., 1999) and a formation factor of 0.85 for the fractures in the large-pore carbonate example. It should be noted that the assumed formation factor of the porous grain domain has significant influences on the estimates of T_2 mode.

We first compare the simulation results against the analytical solution in the sandstone case (i.e., previously shown in Figure 2b). Figure 7 shows an agreement between the analytical T_2 modes (arrow and pin bars) and the simulated T_2 distribution (dashed, solid, and dot-dashed curves) for the fractured sandstone example. Note that T_2 distribution curves are rescaled in amplitude to show the total porosity comparison among each curve; i.e., the area under the coupled curves is equal to summation of areas under the isolated curves. Meanwhile, the amplitude of T_2 modes represents the volume fraction (porosity) of each mode. Therefore, the analytical T_2 modes not only match the T_2 peak positions but also predict the change in volume fraction of each mode.

Next, we compare the numerical simulation results against the analytical model for the large-pore carbonate example (i.e., previously shown in Figure 5). In this case, the role of microfractures and intergranular pores in the 1D analytical model is indeed switched; i.e., the microfractures are the short-time mode, whereas the intergranular pores are the long-time mode. Figure 8 shows the comparison between the analytical T_2 modes (arrow and pin bars) and the simulated T_2 distribution (dashed and solid curves) of the large-pore carbonate rock sample including thin planar fractures. The results demonstrate that analytical T_2 modes are in agreement with the simulated T_2 distribution in the large-pore carbonate case, which further confirms the existence of fracture-pore diffusional coupling.

It should be emphasized that the analytical T_2 bars shown in Figures 7 and 8 are the approximate analytical solutions of the simplified 1D model. Thus, when comparing the T_2 modes from 1D simplified model to the simulation results in 3D volume, the agreement might not be perfect. Ramakrishnan et al. (1999) propose that the accurate solutions to the 1D model can be derived by Laplace transform of equations 5–8, and then final solutions can be obtained by contour integration or numerical Laplace inversion.

DISCUSSIONS

In this paper, we quantified the impact of natural microfractures and the effect of diffusional coupling between microfractures and intergranular pores on NMR response. We considered

Table 6. Parameters assumed in the analytical models for the sandstone and large-pore carbonate examples.

Parameters	Sandstone example (with 9-, 12-, and 15- μm fractures)	Large-pore carbonate example (with 3-, 6-, and 9- μm fractures)
T_{2B} , s	3.1	3.1
D , $\mu\text{m}^2/\text{s}$	2070	2070
ρ , $\mu\text{m}/\text{s}$	10	10
H , μm	4.5, 6, and 7.5 (fractures)	28.5 (pores)
ϕ_G , %	16.51, 16.40, and 16.30 (pores)	9.13, 8.98, and 8.98 (fractures)
$T_{2,G}$, s	0.106	0.107, 0.215, and 0.272
F	25 (intergranular pores)	0.85 (fractures)

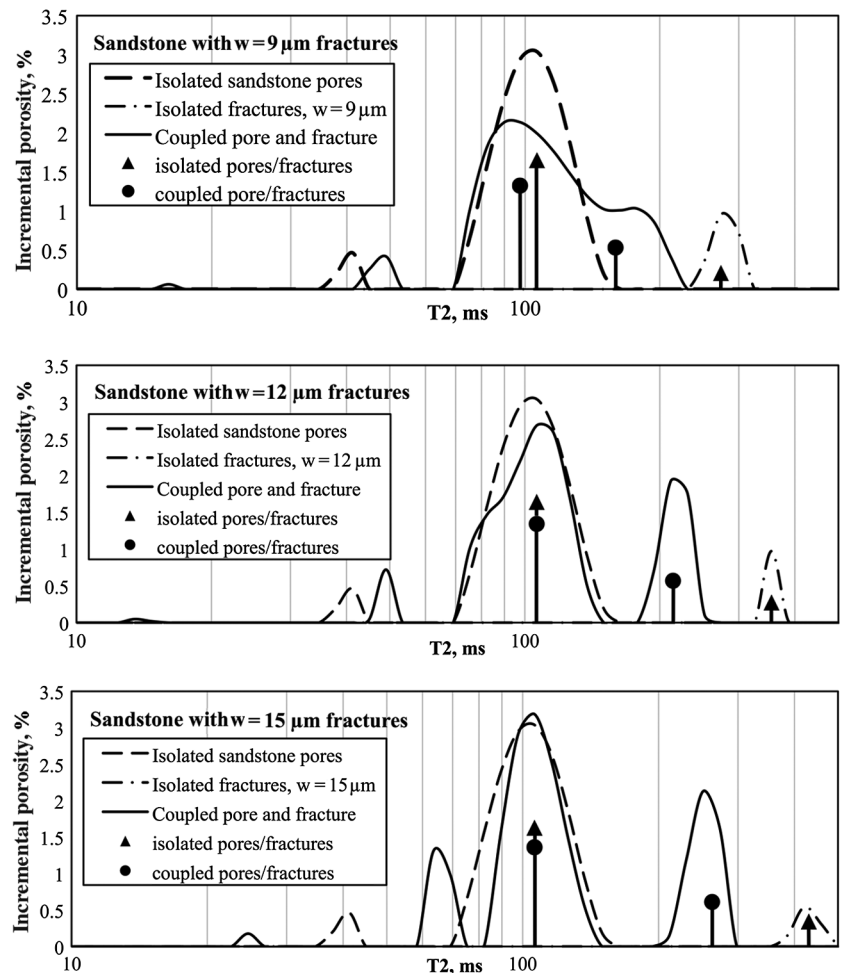


Figure 7. Analytical T_2 modes and simulated T_2 distributions for the sandstone rock sample including planar fractures with width (w) of 9, 12, or 15 μm and length (L) of 270 μm . The arrow bars represent the T_2 modes for isolated sandstone pores or fractures, and the pin bars show coupled fractures or pores. The T_2 curves are the same as in Figure 2b, showing the simulated T_2 distributions for the isolated sandstone pores (dashed), coupled pores-fractures (solid lines), and isolated fractures (dot-dashed), respectively. The area under each T_2 distribution curve represents the total porosity of the rock.

spherical intergranular pores and planar microfractures and channel-like inclusions in our petrophysical models. The results of simulations showed that the effect of diffusional coupling on NMR response is not negligible. When the two forms of porous structures are diffusively coupled, we observed that (1) the longer T_2 component always significantly decreases, and (2) the shorter T_2 component remains unaffected or slightly decreased, and the volume occupied by the shorter T_2 component can be underestimated by NMR T_2 distribution.

We evaluated rock samples with different combinations of pore and fracture sizes. In the cases in which planar fractures are much thinner than one-third of pore size, the pore relaxation times are significantly shortened by the existence of fractures. Thin fractures shift the pore relaxation time more significantly than thick fractures do. In the cases in which planar fractures are much thicker than one-third of pore size, the intergranular pores cause a significant de-

crease in fracture relaxation times. The well-connected pores can shorten the fracture relaxation time more significantly than the poorly connected pores do. Furthermore, the total volume of intergranular pores can be underestimated in this case. Finally, in the cases in which planar fracture apertures close to one-third of pore size, the relaxation times corresponding to microfractures overlap those of intergranular pores. Thus, fractures and pores cannot be distinguished using NMR T_2 distribution in this case.

In all the three aforementioned categories, NMR T_2 distribution is shown to be sensitive to volumetric concentration and shape of microfractures. With higher concentration of fractures, the fracture-pore coupling becomes stronger. Furthermore, planar fractures show more significant impact on T_2 distribution than channel-like inclusions even when they have the same S/V ratio.

All the randomly distributed and oriented microfractures considered in this paper were uniform in width or radius. To investigate the

impact of nonuniform fractures on NMR T_2 distribution, we synthesized planar fractures with variable widths and constant length of fractures in the large-pore carbonate rock sample. Widths of synthetic microfractures were randomly generated based on Gaussian normal distribution. We assumed a mean value of 6 or 9 μm and a 3- μm deviation from mean value. The NMR simulation results (Figure 9) showed that the T_2 distribution in such fractured rocks appears to be similar to those with uniform fractures of 6- or 9- μm width (Figure 5). The pore T_2 of rocks with nonuniform or uniform fractures is the same in particular. The T_2 distribution can, however, get affected by nonuniform fractures in the case of a wide range of fracture thickness. Indeed, in the presence of a wide range of fracture widths, the diffusional coupling between fractures and intergranular pores and among fractures with different widths (if connected) will cause uncertainty in interpretation of NMR response.

In this paper, we differentiate microfractures and pores by aspect ratio and shape. Microfractures refer to pores with low aspect ratio, for example, with aspect ratio of less than 0.1, which is significantly less than that of pores. Some grain contact asperities might also form porous space, and they can be considered as microfractures as long as they are thin and long enough. By such definition, we believe the fracture-pore coupling effect exist in a wide variety of multiple-porosity systems, for example, microfractured grains in tight sands or deformation bands. The theoretical results reported in this paper can be used to explain pore-coupling phenomena in such complicated systems. Furthermore, the outcomes of this paper potentially enable quantifying volumetric concentration of microfractures using NMR, when combined with other well logs such as electromagnetic measurements. Such discrimination between microfractures and pores can facilitate the analysis of pore connectivity and topology, which in turn can enhance interpretation

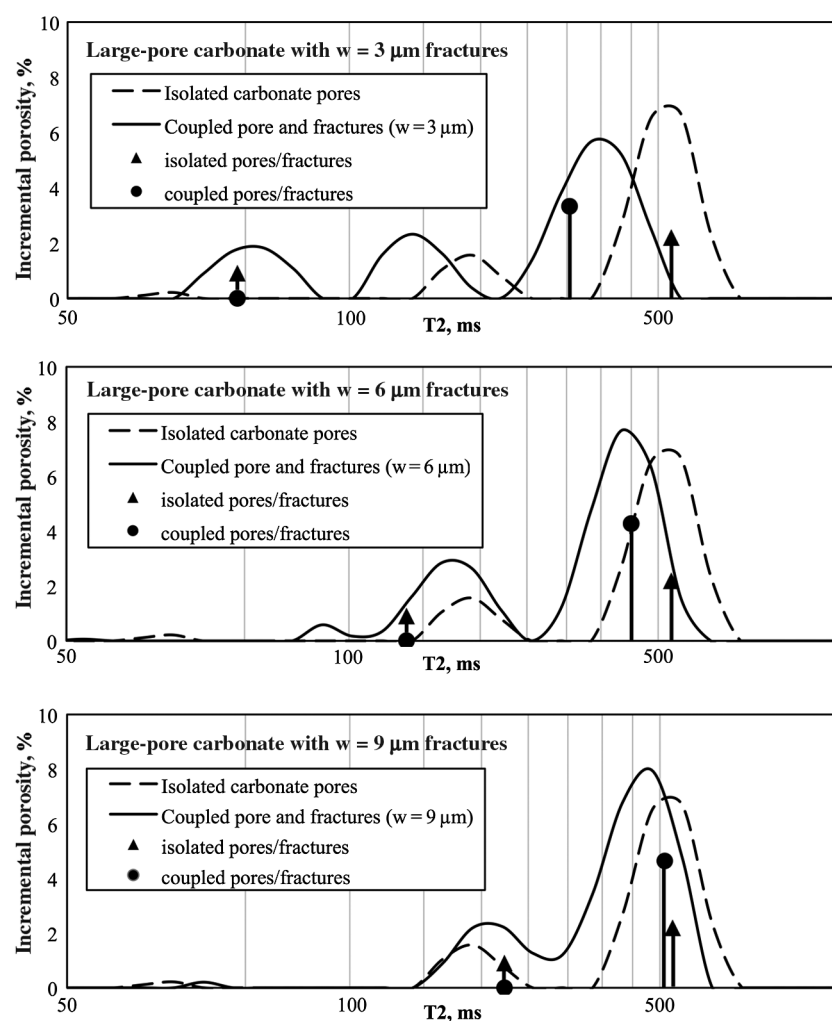


Figure 8. Analytical T_2 modes and simulated T_2 distributions for the carbonate rock sample including planar fractures with width (w) of 3, 6, or 9 μm and length (L) of 150 μm . The arrow bars represent the analytical T_2 modes for isolated carbonate pores or fractures, and the pin bars show coupled fractures or pores. The T_2 curves are same as in Figure 5, showing the simulated T_2 distributions for the isolated carbonate pores (dashed) and coupled pores-fractures (solid lines), respectively. The area under each T_2 distribution curve represents the total porosity of the rock.

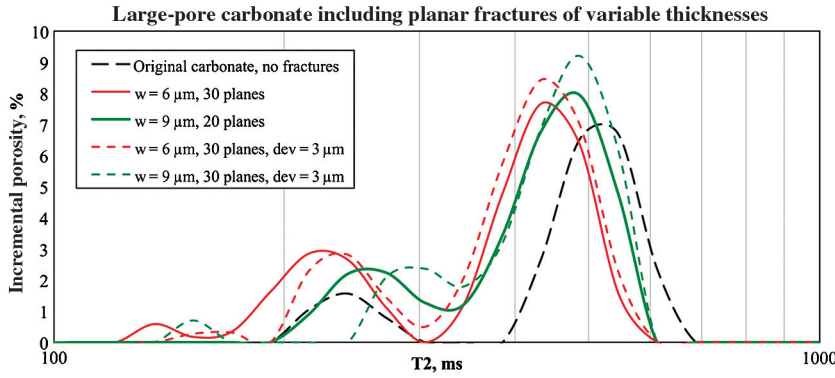


Figure 9. Large-pore carbonate rock sample: Simulated NMR T_2 distribution for the synthetic large-pore carbonate example, in presence of planar fractures of variable widths. The red dashed curve shows the carbonate sample including 30 planar fractures, with a mean width (w) of 6 μm and a 3- μm deviation from the mean value. The green dashed curve shows 20 planar fractures, with a mean width (w) of 9 μm and a 3- μm deviation. They all keep the same length (L) of 150 μm . The red and green solid curves show the same uniform-width planar fractures as in Figure 5. The area under each T_2 distribution curve represents the total porosity of the rock.

of other borehole geophysical measurements such as electromagnetic and acoustic measurements.

CONCLUSIONS

Based on the simulation results and analytical model calculations obtained in this research, and previously published experimental outcomes, we proposed a new concept of fracture-pore coupling and showed that NMR T_2 distribution is influenced by the diffusional coupling between microfractures (or channel-like inclusions) and intergranular pores.

Recognition of the fracture-pore diffusional coupling effect can improve substantially the interpretation of NMR measurements in a variety of rock samples (e.g., naturally fractured carbonates and organic-rich source rocks). It is generally accepted that NMR T_2 distribution represents the pore size distribution in rock samples; however, we demonstrated that the pore size or the pore volume fraction can be significantly underestimated by NMR measurements due to fracture-pore coupling.

The findings of this paper, although a first theoretical step to tackle the complex problems in multiple-porosity systems, have a potential to improve the interpretation of NMR core and borehole measurements in challenging reservoirs. Our findings enable future applications of NMR measurements for the assessment of volumetric concentration of microfractures, in addition to inter-/intragranular pores, when combined with other conventional well logs.

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APPENDIX A

CALCULATION OF NMR RELAXATION TIMES FOR ISOLATED FRACTURES

Theoretically, we can estimate the T_2 value for the planar fractures in Figure 2a by combining equations 1 and 4, which results in

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho_2 \left(\frac{S}{V} \right)_{\text{plane}} = \frac{1}{T_{2B}} + \rho_2 \left(\frac{2}{w} \right)_{\text{plane}}, \quad (\text{A-1})$$

where T_{2B} is 3.1 s, ρ_2 is chosen as 10 $\mu\text{m/s}$, and the thickness of planar fractures w is 9 μm . The resulting T_2 value for the fractures will then be 393 ms. However, our pore-scale simulations, which assume voxelized grain boundary, show a fracture T_2 peak at 274 ms (Figure 2a and Table 1). This difference comes from the discretization of the pore space in the micro-CT images, which overestimates the grain-fluid interface surface area by about 1.5 times (Jin et al., 2009). Equation A-1 can be modified for a "surface-area-corrected" coefficient of 1.5 given by

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho_2 \cdot 1.5 \left(\frac{S}{V} \right)_{\text{plane}} = \frac{1}{T_{2B}} + \rho_2 \cdot \left(\frac{3}{w} \right)_{\text{plane}}, \quad (\text{A-2})$$

which provides T_2 peak location of 274 ms.

The T_2 values for channel-like inclusions can be estimated via

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho_2 \cdot \left(\frac{S}{V} \right)_{\text{cylinder}} = \frac{1}{T_{2B}} + \rho_2 \cdot \left(\frac{4}{d} \right)_{\text{cylinder}}, \quad (\text{A-3})$$

where d is the diameter of the cylindrical channels. Then, we apply a surface-area-corrected coefficient of 1.5 to correct for the discretization effect.

NOMENCLATURE

List of symbols

ρ	= surface relaxivity ($\mu\text{m/s}$)
ϕ_G	= intergranular porosity in the porous grains (%)
d	= diameter of spherical pores or channel-like inclusions (μm)
D	= diffusion coefficient (m^2/s)
F	= formation factor ()

G	= spatial gradient of the internal magnetic field (Hz/mm)
H	= half the thickness of planar fractures (μm)
I_{coupled}	= intensity of coupled microfractures ()
I_{iso}	= intensity of isolated microfractures ()
L	= length of planar fractures or channel-like inclusions (μm)
M_F	= magnetization in fractures per unit volume (A/m^4)
M_G	= magnetization in porous grains per unit volume of pore space (A/m^4)
r	= radius of channel-like inclusions (μm)
T_2	= NMR spin-spin relaxation time (ms)
T_{2B}	= bulk relaxation time (ms)
T_{2S}	= surface relaxation time (ms)
T_{2D}	= diffusion-induced relaxation time (ms)
$T_{2,G}$	= relaxation time in the porous grains (ms)
$T_{2F,\text{coupled}}$	= relaxation time in the coupled microfractures (ms)
$T_{2F,\text{iso}}$	= relaxation time in the isolated microfractures (ms)
TE	= interecho spacing time (μs)
w	= thickness of planar fractures (μm)
W	= distance from fracture center to grain center in the 1D model (μm)
γ	= gyromagnetic ratio of a proton ($\text{rad} \cdot \text{s}^{-1} \cdot \text{Tesla}^{-1}$)

List of acronyms

1D	= one-dimensional
3D	= three-dimensional
CPMG	= Carr-Purcell-Meiboom-Gill
CT	= computed tomography
EPC	= Euler-Poincaré characteristics
FIB-SEM	= focused ion beam scanning electron microscope
NMR	= nuclear magnetic resonance

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